

AS

CONTACT



559-722-2512



asing075@ucr.edu



Los Angeles, CA, 90012



[amninder-singh](#)



[@amnindersahota](#)



[singhamninder](#)



[amnindersahota.com](#)

EDUCATION

Ph.D. | Environmental Sciences (Soil & Water)

UC Riverside
2018 – 2021

MS | Plant Science

California State University | Fresno
2015– 2017

BSc | Agriculture

Punjab Agricultural University |
Ludhiana | Punjab, India
2011 – 2015

LANGUAGES

- English
- Punjabi
- Hindi

AMNINDER SINGH

Geospatial Data Scientist

PROFESSIONAL EXPERIENCE

GEOSPATIAL DATA SCIENTIST (CONTRACT)

Climate | Nov 2022 – Present

Responsible for fulfilling requests for geospatial data analysis from various teams, while also creating and presenting the workflow to stakeholders.

- Contributed to projects on detecting cover crop and tillage practices in agricultural fields using remote sensing.
- Deliver crop type and historical land use information at scale from remote sensing products and extract insights essential to support requests from other teams.
- Collaborated with team members in designing, developing, and testing geospatial pipelines for remote sensing data extraction and machine learning model deployment.
- Evaluated and analyzed spatial indexing systems such as Geohash, H3, and S2 to understand their viability for different use cases. Made recommendations and contributed to the implementation of spatial indexing systems, ensuring they aligned with the team's requirements.
- Used my skills in programming, data manipulation, and visualization to identify data quality issues with spatial datasets and have performed data cleaning to ensure data accuracy and completeness. Used various visualization tools to communicate the findings effectively.

POST-DOCTORAL SCHOLAR

University of California, Riverside & [USDA Salinity Lab](#) | Jan 2022 – Oct 2022

Focus of my work is to study high-resolution soil-plant relationships. Some duties or experiences include -

- Use satellite data from multiple sources, including synthetic aperture radar (SAR) and multispectral imagery for agricultural applications.
- Responsible for gathering large spatial and time-series data using google earth engine's python API and implementing machine learning models for soil moisture estimation using python libraries scikit-learn, TensorFlow, etc.
- Use vegetative indices derived from optical imagery to get phenological parameters of a crop and study relationships with soil salinity/texture.

A M N I N D E R S I N G H

SKILLS

- Programming: Python (pandas, NumPy, matplotlib, scikit-learn, TensorFlow, PySpark etc.), Matlab, and R
- Geospatial: Google Earth Engine, Descartes Labs, ArcGIS, QGIS.
 - Python libraries: xarray, geopandas, shapely, rasterio.
 - Familiar with Microsoft Planetary Computer and interacting with STAC-API using pystac_client.
- Machine Learning and Deep Learning: Proficient in various machine learning algorithms and deep learning frameworks.
- Version control: Git, GitLab, GitHub.
- Soil and environmental sensor network data acquisition systems.
- Geophysical soil sensing.
- Deployment: Django, Streamlit.
- Other: SQL, AWS, Google Cloud Platform.

RELEVANT COURSES

ENSC 175 – Spatial Analysis and Remote Sensing, CS235 – Data mining methods, STAT208 – Statistical Data mining, ENSC 217 – Vadose Zone Processes, ENSC110 – Environmental Statistics, PLANT 251 – Soil Plant Water relations.

PROFESSIONAL EXPERIENCE

- Experience working with other remote sensing datasets for land use and crop type (USDA CDL, USDA NLCD), soil properties (SSURGO, POLARIS), weather (GRIDMET), and SRTM Digital Elevation Data (DEM).
- Proximal sensing of Vegetation Optical Depth (VOD) and soil moisture using portable L-band radiometer in micro-irrigated orchards.
- Experience working with Planet Labs Data and Orders API.
- Collaborating with private companies and data scientists to implement new strategies and fulfill project requirements.

GRADUATE STUDENT RESEARCHER

University of California, Riverside | Jan 2018 – Dec 2021

Dissertation: Advancing Urban Landscape Irrigation Management using Smart Controllers and Machine Learning-Based Models

- Experienced in sampling and analyzing soil hydrology field measurements, including soil water potential, soil volumetric water content, infiltration, evapotranspiration, and weather data.
- Developed machine learning and deep learning models using soil hydrological and time-series weather datasets (ANN, LSTM, and CNNs).
- Collaborated and participated in several field campaigns supporting remote sensing projects.
- Performed data organization, wrangling, and analysis.
- Gained experience with aggregating & statistically analyzing large agricultural datasets using Python, R, or MATLAB.

GRADUATE RESEARCH ASSISTANT

CSU, Fresno | Aug 2015 – Dec 2018

Thesis: Use of EM-38 soil surveys in forage fields at a saline drainage water reuse site to calibrate a hydro-salinity model for decision support.

- Conducted geophysical measurements for soil salinity mapping using electromagnetic sensing platforms.
- Analysis of the data using tools like ArcGIS and R.

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ACHIEVEMENTS AND AWARDS

- Attended IEEE GRSS-USC MHI 2023 Remote Sensing Summer School, from July 13 – 15, 2023, by IEEE Geoscience and Remote Sensing Society, at the University of Southern California (USC).
- Attended Second IADF School on Computer Vision for Earth Observation. IEEE Geoscience and Remote Sensing Society. Oct 3 to Oct 7, 2022. (~12% of the applicants were selected to attend this school).
- Best Oral Presentation Award - 2021 UCR Environmental Sciences Graduate Student Symposium.
- Dissertation Year Program Fellowship, Department of Environmental Sciences, UCR (Fall 2021).
- Hilda and George Liebig Environmental Sciences Summer Fellowship, Department of Environmental Sciences, UCR (2020).
- SCSC-NWRI Fellowship, Southern California Salinity Coalition (SCSC), and the National Water Research Institute (NWRI), 2018-2020.
- Stolzy-Letey Travel Scholarship, Department of Environmental Sciences, UCR (2018-2019).
- Mini GSA Award, Hilda and George Liebig Foundation, Department of Environmental Sciences, UCR (2019, 2021).
- UCANR Drone Camp, June 18-20, 2018, UC San Diego
- Outstanding Thesis Award. Jordan College of Agricultural Sciences & Technology, CSU, Fresno (2017-2018 academic year).
- Graduate Student Achiever, Department of Plant Science, CSU, Fresno. (2018).
- Ag One-John P. "Phil" Larson Scholarship, CSU, Fresno (Spring 2017).
- Inducted to The Honor Society of Phi Kappa Phi (Spring 2017).
- Jim Pattersen's Ag Science student recognition award - Agriculture Science Student of the Year, 23rd Assembly District, (2017).
- The Gerald O. Mott Meritorious Graduate Student Award in Crop Science, Crop Science Society of America (CSSA), (2017).
- Third place in Poster competition, Plant and soil conference – American Society of Agronomy, California chapter, (2017).
- Jordan Assistantship, Jordan College of Agricultural Sciences and Technology, CSU, Fresno (August 2015-May 2017).
- Graduate Research Fellowship, Graduate Net Initiative, CSU, Fresno (2016-2017).
- Travel grant, Division of Graduate Studies, CSU, Fresno for the 2016 ASA, CSSA, and SSSA Annual Meeting, November 5-9, 2016, in Phoenix, AZ.
- Dean's Scholarship Tuition Waiver, CSU, Fresno (2015- 2016).
- Punjab State Marketing Board Scholarship, PAU, Ludhiana (2011 - 2015).

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PUBLICATIONS

- **Singh, A.**, Haghverdi, A., 2023. Development and evaluation of temperature-based deep learning models to estimate reference evapotranspiration. Artificial Intelligence in Agriculture. <https://doi.org/10.1016/j.aiia.2023.08.003>
- Sapkota, A., Haghverdi, A., Merhaut, D., **Singh, A.**, & Iradukunda, J. C., 2023. Response of Landscape Groundcovers to Deficit Irrigation: An Assessment Based on Normalized Difference Vegetation Index and Visual Quality Rating. HortScience, 58(3), 274-285. Available at <https://doi.org/10.21273/HORTSCI16915-22>.
- Haghverdi, A., Reiter, M., **Singh, A.**, and Sapkota, A. 2021. Hybrid Bermudagrass and Tall Fescue Turfgrass Irrigation in Central California: II. Assessment of NDVI, CWSI, and Canopy Temperature Dynamics. Agronomy 11(9): 1733 Available at <https://www.mdpi.com/2073-4395/11/9/1733>.
- Haghverdi, A., Reiter, M., Sapkota, A., and **Singh, A.** 2021. Hybrid Bermudagrass and Tall Fescue Turfgrass Irrigation in Central California: I. Assessment of Visual Quality, Soil Moisture and Performance of an ET-Based Smart Controller. Agronomy 11(8): 1666. Available at <https://doi.org/10.3390/agronomy11081666>.
- **Singh, A.**, Haghverdi, A., Öztürk, H.S., and Durner, W. 2021. Developing Pseudo Continuous Pedotransfer Functions for International Soils Measured with the Evaporation Method and the HYPROP System: II. The Soil Hydraulic Conductivity Curve. Water 13(6): 878 Available at <https://www.mdpi.com/2073-4441/13/6/878>.
- Haghverdi, A., **Singh, A.**, Sapkota, A., Reiter, M., and Ghodsi, S. 2021. Developing irrigation water conservation strategies for hybrid bermudagrass using an evapotranspiration-based smart irrigation controller in inland southern California. Agric. Water Manag. 245(April 2020): 106586 Available at <https://doi.org/10.1016/j.agwat.2020.106586>.
- **Singh, A.**, Haghverdi, A., Öztürk, H.S., and Durner, W. 2020. Developing pseudo continuous pedotransfer functions for international soils measured with the evaporation method and the hyprop system: I. the soil water retention curve. Water (Switzerland) 12(12): 1–17 Available at <https://www.mdpi.com/2073-4441/12/12/3425>.
- **Singh, A.**, Quinn, N.W.T., Benes, S.E., and Cassel, F. 2020. Policy-Driven Sustainable Saline Drainage Disposal and Forage Production in the Western San Joaquin Valley of California. Sustain. 2020, Vol. 12, Page 6362 12(16): 6362 Available at <https://www.mdpi.com/2071-1050/12/16/6362/htm>
- **Singh, A.**, Haghverdi, A., Nemati, M., and Hartin, J. 2020. Efficient Urban Water Management Smart Weather-Based Irrigation Controllers. Univ. Calif. Agric. Nat. Resour. 8674(July): 1–11